

Overview

Introduction

Hello and welcome to Module 6 of COMP-304. In this module, you will learn about Networking and Accessing Internet Resources in Android Apps.

Learning Outcomes

At the end of this module, you will be able to:

1. Discuss the principles of asynchronous programming and Kotlin coroutines.
2. Set up Retrofit for making network calls in Android applications.
3. Analyze the basics of Kotlin coroutines and their benefits in handling asynchronous tasks.
4. Implement network calls using Kotlin coroutines and Retrofit.

Checklist

There are several tasks you'll need to complete:

1. Review the online content for this module, which includes reviewing the module's key terms and concepts.
2. Run all class examples.
3. Complete the midterm practice session.
4. Submit and Demonstrate Lab Assignment 2

Key Terms and Concepts

Here are a few terms you need to be familiar with.

Term	Definition
Kotlin Coroutines	A coroutine is a concurrency design pattern that you can use on Android to simplify code that executes asynchronously.
Flow	A flow is a type that can emit multiple values sequentially, conceptually a stream of data that can be computed asynchronously. Flows are built on top of coroutines and can provide multiple values.
Retrofit	Retrofit - the de facto official way to communicate with an HTTP API on Android
Moshi	Moshi (github.com/square/moshi) is another library from Square. Moshi maps JSON data to Kotlin objects automatically.
Coil	Coil is used for efficient image loading.
HttpURLConnection	A Java class used for sending and receiving data over the web, allowing for HTTP requests handling in a low-level, manual way.
OkHttp	An efficient HTTP client by Square, used for managing network requests either as a standalone client or as the engine for Retrofit.
Volley	A networking library developed by Google that simplifies network operations and manages network requests queue automatically
Gson	A library used to convert Java/Kotlin objects into their JSON representation and vice versa, often used with Retrofit for automatic data parsing.
JSON	A lightweight data interchange format that is easy for humans to read and write, and easy for machines to parse and generate.

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Summary

Key Take-Aways

Here are a few key take-aways from this module.

- We introduced Kotlin coroutines to manage long-running tasks.
- We introduced Flow as an alternative to LiveData, for emitting multiple values sequentially.
- We connected to internet resources and implemented parsing by using Retrofit, Coil, and Moshi.

Don't forget to complete the items on your checklist.

Summary Checklist

Don't forget to complete the items on your checklist:

1. Review the online content for this module, which includes reviewing the module's key terms and concepts.
2. Complete the hands-on midterm test during the scheduled session.
3. Read articles/tutorials given in Learning Activities.

Works Cited

1. Textbook, Chapter 20
2. <https://kotlinlang.org/docs/async-programming.html#coroutines>
3. <https://developer.android.com/codelabs/kotlin-coroutines#1>
4. <https://developer.android.com/codelabs/advanced-kotlin-coroutines#1>
5. <https://developer.android.com/kotlin/flow>
6. <https://developer.android.com/topic/libraries/architecture/livedata#java>
7. <https://developer.android.com/topic/libraries/architecture/livedata#java>

Coming up next, we'll be looking at the weekly topic for module 7 as identified in the approved course outline.

Works Cited

1. Textbook, Chapter 20
2. <https://kotlinlang.org/docs/async-programming.html#coroutines>
3. <https://developer.android.com/codelabs/kotlin-coroutines#1>
4. <https://developer.android.com/codelabs/advanced-kotlin-coroutines#1>
5. <https://developer.android.com/kotlin/flow>
6. <https://developer.android.com/topic/libraries/architecture/livedata#java>
7. <https://developer.android.com/topic/libraries/architecture/livedata#java>

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